A physics-exact credit decomposition for distributed flutter suppression, and the exploration scale that hides it

Quang-Vinh Dang[a], Hai Dang Nguyen[b], Ngoc Minh La[c], Minh Ngoc Dinh[d,*]

[a] *British University Vietnam, Hung Yen, Vietnam*
[b] *XTECH AI Technology, Ho Chi Minh City, Vietnam*
[c] *Vietnamese German University, Ho Chi Minh City, Vietnam*
[d] *School of Data Science & Information Technology, Millennia University, Ho Chi Minh City, Vietnam*

Abstract

Differentiating the aeroelastic energy of a structural plant along its trajectories isolates a control-power term exactly additive across an arbitrary number of control inputs. We use this to give multi-agent reinforcement learning, applied to distributed flutter suppression on several independently actuated surfaces of a Goland wing cross-validated against an unsteady vortex-lattice solver, a closed-form per-agent credit signal in place of the estimated one the method normally requires. Combined with log-energy shaping, this credit beats the closed form alone ($g = -2.49$, corrected $p = 0.020$, also against a naive reward and an adapted VDN-style baseline); it does not significantly beat shaping alone at that standard, which a less rigorous earlier version overstated, and we report the retraction with the finding. Its advantage over the estimated alternatives is reliability, not raw mean: both match or exceed it on some seeds and fail on others, at higher variance.

None of this was visible at conventional exploration settings. A controller achieving the classical result uses 0.105° RMS deflection, against 4.5° for a conventional exploration standard deviation; every mechanism we tried plateaued until exploration was matched to that amplitude, changing performance by a factor of two and making the credit result measurable at all. We further checked that the credit's sign tolerates plausible identification error and that the classical baseline's envelope-edge divergence is not simply

*Corresponding author.

Email addresses: vinh.dq4@buv.edu.vn (Quang-Vinh Dang), haidang5101994@gmail.com (Hai Dang Nguyen), lmngocdhsp@gmail.com (Ngoc Minh La), minh.dinh@maeducation.com (Minh Ngoc Dinh)

a tuning artefact. The learned policies still trail a decentralised LQG on average damping (-1.53 against $-2.05\,\mathrm{s}^{-1}$) while holding one condition it cannot, pointing toward a certified-plus-monitored-residual architecture we have not yet built.

Keywords: active flutter suppression, aeroelasticity, multi-agent reinforcement learning, credit assignment, decentralised control, exploration

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Nomenclature

$q, \dot{q}, \ddot{q}$	modal coordinates, rates, accelerations
M, C, K	modal mass, damping, stiffness matrices
L, x_{w}	wake-coupling matrix and wake (Wagner lag) states
B, δ, b_k	control-influence matrix, deflection vector, its kth column
E	instantaneous structural energy, $\frac{1}{2}\dot{q}^{\top}M\dot{q} + \frac{1}{2}q^{\top}Kq$
P_k	control power of agent k, $(\dot{q}^{\top}b_k)\delta_k$
λ	energy decay rate, $\frac{1}{2}\,\mathrm{d}(\log E)/\mathrm{d}t$
D_k	difference reward for agent k
U, U_{f}	airspeed, flutter speed
N	number of control surfaces / agents
R	number of retained modal phasors (R=2 throughout)
o_k, a_k	agent k's local observation and normalised action
$\hat{\phi}_k$	agent k's encoded phasor estimate, $\in \mathbb{R}^{2R}$
e_k	consensus message received by agent k
σ	initial policy exploration standard deviation (normalised action units)
d, p	Cohen's d effect size, Welch's t-test p-value

1. Introduction

Flutter is a self-excited aeroelastic instability in which structural bending and torsion couple through unsteady aerodynamic loads and draw energy from the freestream. Past a critical speed the coupled mode becomes unstable and the response grows without bound, typically within a few oscillation cycles and with no warning a pilot can act on. It has been a design driver since the earliest flexible aircraft and remains one: because a wing must be certified against flutter with a substantial speed margin, the mass, stiffness and manufacturing cost of the structure are set in large part by an instability the aircraft is never meant to encounter in service (Livne, 2018;

Mukhopadhyay, 2003). Active suppression offers a way to buy back some of that margin with control effort rather than with structural weight, which is why it has attracted sustained engineering attention since the 1970s (Karpel, 1982; Waszak, 2001; Block and Strganac, 1998).

Two trends motivate revisiting the problem now. First, wings are becoming more flexible — for fuel efficiency, span, and load alleviation — and flexible wings carry more independently actuated trailing-edge surfaces, each with its own local sensing and local computation on a bus rather than a dedicated wire to a central controller. The hardware is already distributed even where the control law is not. Second, reinforcement learning has been applied to flutter suppression with encouraging results: circulation control validated in a wind tunnel (cir, 2023), camber morphing for stall flutter (of Fluids, 2025), and tuning the parameters of a conventional controller (fly, 2022). Every instance we are aware of, however, uses a single agent commanding a single effector from a global observation. Taking the hardware trend seriously means asking what happens when the effectors are commanded by separate agents that cannot see the whole wing, which is the question this paper answers on a specific, validated testbed (Figure 1).

Posing the problem with one agent per surface makes credit assignment the central difficulty, and for a physical reason rather than an algorithmic one: every surface acts on the *same* coupled bending–torsion mode, so a reward built from the whole wing's response says almost nothing about which surface actually helped. The standard remedies in multi-agent reinforcement learning estimate that decomposition statistically, from the returns a policy happens to collect (Sunehag et al., 2018; Rashid et al., 2018; Foerster et al., 2018). Our first observation is that this particular class of plant does not require estimation at all: for a structural system whose energy budget is exactly additive across control inputs, the credit decomposition is available in closed form from the equations of motion themselves, with no learning involved in computing it.

Our second observation is the one we would most want a practitioner to take away, and we did not anticipate it going in. Before finding it, we spent several rounds comparing credit signals, communication schemes, action parameterisations, recurrence and auxiliary losses — the ordinary methodology of an ablation study — and every variant plateaued in the same narrow performance band regardless of which mechanism it used. That invariance was not evidence that the mechanisms were equivalent to one another; on closer inspection it was evidence that something *upstream* of all of them was binding. The exploration standard deviation of the policy was roughly forty times the deflection amplitude that actually damps this wing, so no policy ever experienced the fine, precisely-phased actuation that works, and no credit signal — however well constructed — could assign credit to a signal buried that far under noise. Matching the exploration scale to the useful actuation amplitude changed performance by a factor of two, more than any single algorithmic choice we tried, and only after that correction did the credit-assignment effect we had set out to study become measurable at all.

This is a known failure mode in spirit: the reinforcement learning literature has repeatedly found implementation and hyperparameter choices dominating algorithmic ones in general-purpose control benchmarks (Henderson et al., 2018; Engstrom et al., 2020; Andrychowicz et al., 2021). Our contribution is a control-specific instance of that caution, with a diagnostic a practitioner can use *before* training anything: the useful actuation amplitude is computable from a classical design, and the exploration scale can be checked against it in advance rather than discovered by trial and error.

Contributions.

1. A closed form for the multi-agent difference reward in an energy-conserving structural plant with additive control authority (Proposition 1), together with the correction that makes it usable: the credit must be the logarithmic decay rate, not the raw energy rate (Section 4.2).
2. A five-variant credit-assignment comparison, corrected for the number of comparisons made, against a naive undecomposed reward and an adapted value-decomposition baseline in the style of VDN as well as against each half of the closed-form-plus-shaping credit alone: the closed-form credit's advantage over an estimated one is in consistency across seeds and conditions, not a uniformly larger effect size (Section 5.7).
3. A quantified account of what limits learning on this problem, with a diagnostic a practitioner can apply in advance, and a capability probe that rules out the competing explanation (Section 5.9), together with a direct check of how much control-influence identification error the credit signal's sign tolerates (Section 5.10).
4. A validated testbed and a six-rung classical baseline ladder that separates the control method from the information structure available to it, cross-validated against an unsteady vortex-lattice solver, including a check of whether the classical baseline's fragility at the envelope edge is a tuning artefact rather than an assumption (Sections 3.3, 5 and 5.6).
5. A negative result reported as such: the communication and action parameterisation mechanisms we derived from the physics do not help,

at a level of statistical correction that makes the negative result harder to dismiss rather than easier.

2. Related work

Active flutter suppression. The classical approach synthesises a single centralised regulator from a state-space aeroelastic model, driving all control surfaces from one observer and one gain (Karpel, 1982). Robust multivariable designs following that template have been demonstrated on wind-tunnel models (Waszak, 2001), and the approach extends to nonlinear aeroelastic structures with limit-cycle behaviour (Block and Strganac, 1998); Livne (Livne, 2018) surveys the state of the art and the gap between laboratory demonstrations and certified hardware, and Mukhopadhyay (Mukhopadhyay, 2003) traces the field's history from Theodorsen's original frequency-domain analysis to modern aeroservoelastic synthesis. Two things stand out reading this literature together. First, every design in it is re-derived for a new airframe: the state-space model, the actuator suite and the sensor placement all change with the aircraft, and Livne's survey is explicit that the gap between a laboratory demonstration and certified hardware is as much about this re-engineering cost as about control-theoretic novelty. Second, and more consequential for this paper, none of these treatments questions the centralised structure itself: the regulator sees the whole state and commands every surface jointly, which is exactly the assumption a physically distributed set of actuators — multiple independently wired trailing-edge surfaces on a modern flexible wing, each with its own local sensing — need not satisfy. Karpel's original formulation (Karpel, 1982) already treats gust alleviation and flutter suppression as the same state-feedback problem, and we inherit that framing for the objective in Section 3 even though the control structure we study departs from it. The unsteady aerodynamic models underlying both the classical literature and our own plant descend from Theodorsen's frequency-domain analysis (Theodorsen, 1935) and the indicial (Wagner-function) formulation of Jones (Jones, 1940); finite-state induced-flow models (Peters et al., 1995) are the modern sectional alternative, extending the same lag-state idea to a richer wake representation than the two-state approximation we use, and (Fung, 1955; Bisplinghoff et al., 1955; Palacios and Cesnik, 2023) are the standard texts we draw the modelling conventions from, with Palacios and Cesnik (Palacios and Cesnik, 2023) in particular covering the coupled flight-mechanics setting our plant deliberately simplifies away from.

Adaptive flutter suppression. Between the fixed-gain classical designs above and the learned policies of the next paragraph sits a mature adaptive-control literature aimed at exactly the gap our own motivation cites — an aerodynamic model that is never exactly the design model in flight. Downs and Prazenica (Downs and Prazenica, 2023) apply direct adaptive control to a three-dimensional flexible wing with nonlinear stiffness and actuator freeplay, stabilising post-flutter conditions a fixed-gain LQR cannot; Mozaffari-Jovin et al. (Mozaffari-Jovin et al., 2024) design an $\mathcal{L}_1$ adaptive law for a flapped aeroelastic section under unmatched nonlinear uncertainty, the same flap-actuated setting we study. Both retain a single, centralised regulator per airframe and adapt its parameters online rather than decomposing the effectors into separate agents, so they answer a different question from the one this paper asks — robustness to model error at one control point, rather than what changes when the control points themselves are separate decision-makers — and are a closer classical relative to our problem than the fixed-gain designs above, worth situating against directly rather than folding into the same paragraph as work that does not address model uncertainty at all.

Learning for flutter control. A recent line of work applies deep reinforcement learning directly to flutter suppression: stall flutter via active camber morphing (of Fluids, 2025), trailing-edge circulation control validated in a wind tunnel (cir, 2023), and tuning the parameters of a conventional aeroservoelastic controller (fly, 2022). The appeal of learning here is specific to the problem, not generic: flutter is a nonlinear instability once the response leaves the small-perturbation regime the classical model is valid in, the aerodynamic model available in flight is never exactly the design model, and a policy that has seen a distribution of conditions in simulation can in principle interpolate across off-nominal cases a fixed gain cannot. Despite using different effectors and different plants, the three studies above share the same structural assumption as the classical literature they replace a hand-designed gain in: one learning agent, one effector, one global observation. None asks what changes when the effectors are commanded by separate agents that cannot see the whole wing, which is the question a wing with several independently actuated surfaces on a shared bus makes practically relevant and is the gap this paper occupies. Framing a wing's own surface as a multi-agent system is not without precedent: Shalymov et al. (Shalymov et al., 2020) cover an entire wing in many small independently actuated "feathers" and synthesise a speed-gradient adaptive control law across them, which is multi-agent *control* in the classical sense. What we are not aware of is prior work that trains separate *learning* agents, each with only its own local, partial observation of the plant, to solve this problem — the distinction that matters here, since the entire credit-assignment question this paper poses (Section 2, "Multi-agent credit assignment") only arises once each surface's controller is estimated from experience under partial observability rather than derived, as in (Shalymov et al., 2020), from a single control law with access to the whole system's dynamics.

Decentralised control. That decentralisation costs closed-loop performance relative to a centralised design with the same model is a classical result, not a symptom of using learning (Šiljak, 1991; Bakule, 2008). Bakule's survey (Bakule, 2008) catalogues exactly the design patterns this paper's baseline ladder instantiates — decentralised state feedback with no inter-subsystem model, and decentralised observers each estimating only a local state — and reports the same qualitative cost of decentralisation across a wide range of application domains, well outside aeroelasticity. The structural conditions under which decentralised synthesis remains *tractable* at all, rather than merely stable, are characterised by Rotkowitz and Lall (Rotkowitz and Lall, 2006): convex synthesis survives decentralisation exactly when the information structure is *quadratically invariant* with respect to the plant, a condition our line-graph sensing pattern does not satisfy in general, which is part of why the fully decentralised LQG of Appendix D is synthesised independently per surface rather than as a single convex program. This literature is why we treat the information structure as an experimental variable rather than an afterthought: comparing a distributed learned policy against a centralised regulator conflates the control method with the information available to it, and the honest classical baseline for a policy that sees only local accelerometers is a fully decentralised LQG built from the same sensing, not the centralised design that the learned policy could never in principle match.

Multi-agent credit assignment. The standard tools for the credit-assignment problem we pose are difference rewards (Wolpert and Tumer, 2001) and learned value factorisation (Sunehag et al., 2018; Rashid et al., 2018), with COMA (Foerster et al., 2018) using a counterfactual advantage baseline to the same end; MADDPG (Lowe et al., 2017) and MAPPO (Yu et al., 2022) are the common actor-critic baselines built on top of these ideas, the field more broadly is surveyed in (Albrecht et al., 2023; sca, 2025), and estimators continue to be refined (mul, 2025). These methods differ in how much they assume about the environment, but all of them estimate the credit decomposition statistically, from returns a policy happens to collect: VDN and

QMIX (Sunehag et al., 2018; Rashid et al., 2018) learn a factorisation of a joint value function, COMA (Foerster et al., 2018) learns a counterfactual baseline by marginalising a learned critic over one agent's action, and the original difference-reward formulation (Wolpert and Tumer, 2001) itself typically requires a counterfactual rollout with the agent of interest removed. All of this machinery exists *because* the environment is treated as a black box. Our contribution is narrower and complementary rather than competing with this literature: for the specific class of energy-conserving structural plant studied here, the decomposition these methods estimate is available in closed form directly from the equations of motion (Proposition 1), with no learning, no counterfactual rollout and no critic involved in computing it. We see this less as a competitor to the general-purpose methods above than as evidence that physical structure, where it exists, is worth exploiting before reaching for a general estimator built to work without it. The formal setting for the distributed problem itself is the decentralised POMDP (Bernstein et al., 2002; Oliehoek and Amato, 2016), whose complexity results motivate treating partial observability as intrinsic to the sensing architecture (Section 3) rather than as a modelling simplification to be relaxed.

Learned communication. CommNet (Sukhbaatar et al., 2016) and TarMAC (Das et al., 2019) learn message content end to end from a broad channel, with TarMAC additionally learning *who* to attend to via a signature-based attention mechanism over neighbours. We took the opposite design choice on both counts: message *content* is fixed to a modal phasor derived from the physics rather than learned end to end, and message *routing* is fixed to the spanwise line graph rather than learned, on the reasoning that this would make the channel interpretable and bound its bandwidth independently of the number of agents — a property that matters on a physical avionics bus in a way it does not in the simulated coordination games this literature is usually evaluated on. Measured against a plain policy with no channel at all, neither the fixed content nor the fixed routing helped on this plant (Section 5.7), and we report the negative result rather than omit the mechanism; Section 6 returns to what we think this does and does not say about learned communication more broadly.

Multi-agent benchmarks and physical grounding. The environment in this paper is implemented against the PettingZoo parallel-environment API (Terry et al., 2021), the standard interface much of the multi-agent reinforcement learning literature above is evaluated through, including the particle-world benchmarks introduced alongside MADDPG (Lowe et al., 2017). That shared interface is deliberate: it is what lets the credit-assignment and communication mechanisms surveyed above be dropped into this environment largely unmodified, which is what makes the ablations of Section 5.7 a fair test of those mechanisms rather than a test of our own re-implementation of them. Where this environment differs from the particle-world and grid-world benchmarks that dominate the credit-assignment literature is that its dynamics are not designed for the coordination problem they pose; they are a validated model of a physical instability (Section 3.3), and the credit structure in Proposition 1 is a property of that physics rather than of the environment's authorial choices. We think this is a meaningful axis along which physically-grounded multi-agent control tasks are currently under-represented relative to abstract coordination games, and offer this testbed as one instance of closing that gap rather than as a claim that abstract benchmarks are not useful for what they are designed to measure.

Reinforcement learning practice. Our central caution has direct precedent in general-purpose reinforcement learning, though not, to our knowledge, applied to a physical control problem where the binding quantity can be predicted in advance. Henderson et al. (Henderson et al., 2018) documented how seed-to-seed variance can swamp the algorithmic differences a paper sets out to measure; Engstrom et al. (Engstrom et al., 2020) showed that implementation details, not the nominal algorithm, accounted for much of PPO's reported advantage over TRPO; Andrychowicz et al. (Andrychowicz et al., 2021) found hyperparameters dominating architectural choices at scale; and Agarwal et al. (Agarwal et al., 2021) argued for reporting effect sizes and uncertainty rather than point estimates from few runs, a practice we follow throughout Section 5.7. Yu et al. (Yu et al., 2022) make a related point specific to the multi-agent setting: much of MAPPO's reported advantage over more elaborate multi-agent actor-critic methods traces to implementation details — value normalisation, advantage standardisation, the choice of critic input — rather than to the algorithm's nominal novelty, which is the multi-agent analogue of Engstrom et al.'s finding and part of why Section 4.4 documents our own architecture choices at the level of individual design decisions rather than only by name. What we add to this literature is a case in which the dominant hyperparameter — the exploration scale — is not merely discoverable by sweeping it, but *computable in advance* from a classical design of the same plant, which is not true of hyperparameters in the general-purpose benchmarks this literature studies. Residual formulations, in which a learned policy corrects a fixed classical controller rather than replacing it (Johannink et al., 2019; Silver et al., 2018), appear here as

Table 1: This paper against the literature it draws on, along the axes Section 2 discusses. A ✓marks that the property holds for the representative citation(s) in that row; `-` marks that it does not; `n/a` marks that the axis does not apply to that line of work at all (a single-agent or single-regulator method has no multi-agent credit decomposition to be closed-form *or* estimated, so neither ✓nor `-` would be meaningful there).

	Multi-agent surfaces	Closed-form credit	Validated against higher-fidelity solver
Classical AFS (Karpel, 1982; Waszak, 2001)	–	n/a	varies
Adaptive AFS (Downs and Prazenica, 2023; Mozaffari-Jovin et al., 2024)	–	n/a	–
Learned AFS (of Fluids, 2025; cir, 2023; fly, 2022)	–	–	–
General MARL credit (Sunehag et al., 2018; Rashid et al., 2018; Foerster et al., 2018)	✓	–	n/a
Residual control (Johannink et al., 2019; Silver et al., 2018)	–	–	n/a
This paper	✓	✓	✓

one of the variants we evaluate rather than as the proposed method.

Positioning. Table 1 summarises the four axes above together, since no single prior study we are aware of combines more than one of them: multi-agent treatment of the individual control surfaces of one wing, a credit signal derived in closed form from the plant's own physics rather than estimated, validation of the plant itself against a higher-fidelity solver rather than only against a published linear benchmark, and an explicit, reproducible account of what a conventional exploration setting hides. Each axis is individually present somewhere in the literature surveyed above; their combination, and the negative and cautionary results that combination surfaces, is what this paper contributes.

3. Problem formulation and plant model

3.1. Decentralised partially observable decision process

We model the task as a Dec-POMDP (Bernstein et al., 2002; Oliehoek and Amato, 2016) $\langle \mathcal{I}, \mathcal{S}, \{\mathcal{A}_k\}, T, \{\Omega_k\}, O, R, \gamma \rangle$. The agent set $\mathcal{I} = \{1, \ldots, N\}$ has one member per control surface. The state $s = (q, \dot{q}, x_{\mathrm{w}}, \delta)$ collects the modal coordinates, their rates, the wake states and the current deflections. Agent k chooses a normalised command $a_k \in [-1, 1]$, which its actuator converts to a physical deflection subject to a rate limit and travel stops.

Each agent observes only $o_k \in \Omega_k$: an accelerometer pair at its own station, local plunge rate, twist and twist rate, its own saturation fraction, and air data. Crucially o_k does not contain the modal state. The unstable mode is a global quantity that no single station can measure, so the partial observability is intrinsic to the physical arrangement rather than an artefact of our formulation. Communication, where used, is nearest-neighbour along the span.

3.2. Objective

We use the standard discounted formulation (Sutton and Barto, 2018). The quantity to be minimised is the exponential growth rate of the structural energy E, which is also the quantity we report:

$$\lambda \;=\; \tfrac{1}{2}\,\frac{\mathrm{d}}{\mathrm{d}t}\log E, \tag{1}$$

negative when the wing is being damped. Stating the objective this way, rather than as an accumulated quadratic cost, matters later: it is what allows the per-agent credit and the evaluation metric to be literally the same quantity (Section 4.2).

3.3. Model

We use a modally reduced uniform cantilever wing, shown schematically in Figure 1. Three trailing-edge surfaces occupy spanwise bands at $[0.05, 0.40]$, $[0.40, 0.72]$ and $[0.72, 0.98]$ of the semispan (panel a), hinged at 75 % chord (panel b), with $\pm 15^\circ$ travel, $200\,^\circ\,\mathrm{s}^{-1}$ rate limit and a 20 ms first-order servo lag. Each surface is commanded by its own agent, which observes only a local accelerometer pair, local plunge and twist rate, its own saturation fraction, and air data; no agent observes the modal state, and the three agents share no channel unless a communication mechanism is explicitly under test (panel c). Control runs at 200 Hz.

3.4. What flutter and its suppression look like

Every quantity in the rest of the paper is a scalar derived from a wing that is physically bending and twisting, and it is worth seeing that motion once before reducing it to decay rates. Figure 2 gives the same initial 5 cm tip pluck two treatments at $U/U_\mathrm{f} = 1.10$, six shared instants apart, on one

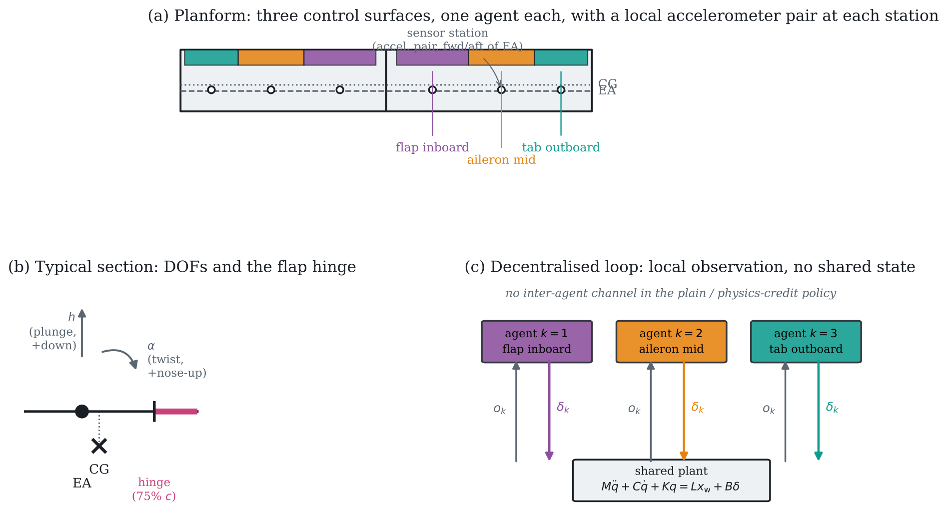

Figure 1: The testbed. (a) Planform, agent colours matched throughout the paper. (b) The typical-section degrees of freedom and the flap hinge that the thin-airfoil derivative in Section 3.6 refers to. (c) The information structure: each agent closes its own loop through the shared plant, with no inter-agent channel in the plain and physics-credit policies.

shared deformation scale so a flat-looking row is not an axis artefact. Left to right along the top row is exactly what "flutter" names: bending and torsion lock into a fixed phase relationship and the response grows, visibly, within under a second, until the linear model is no longer valid. The bottom row is the identical pluck under centralised LQR: by the second frame the deflection is already smaller than it was at $t = 0$, and it stays there. Neither row is a simulation artefact of the illustration — both are literal rollouts of the plant in Section 3.3 at the flight condition used throughout Section 5.5, rendered rather than tabulated. A supplementary animation of all four cases in the classical ladder (open loop, centralised LQR, local rate feedback, and centralised LQR with the outboard surface jammed) is provided alongside the code (Code and data availability); a static figure necessarily samples six instants; the animation shows the coalescing bending–torsion cycle itself, which the sampling interval here is deliberately too coarse to resolve.

The structure is a Galerkin projection onto exact clamped-free bending and fixed-free torsion modes, with all generalised mass and stiffness integrals evaluated by Gauss–Legendre quadrature. Aerodynamics are strip theory carrying the full Theodorsen non-circulatory terms (Theodorsen, 1935), with the circulatory lag realised in the time domain by two Wagner lag states per

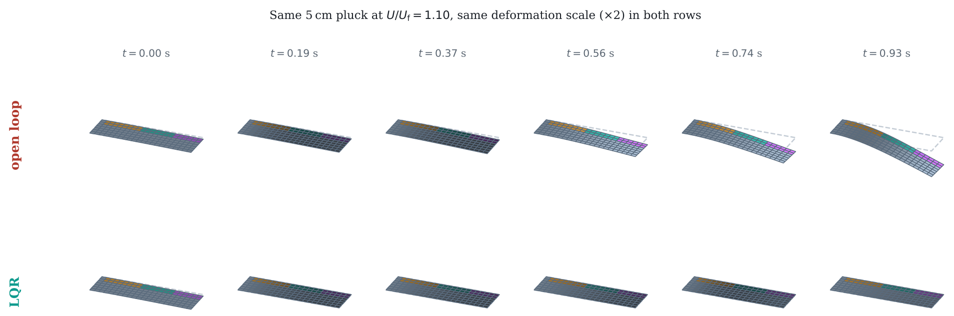

Figure 2: Six shared instants of the same 5 cm initial tip pluck at $U/U_{\mathrm{f}} = 1.10$, one shared deformation scale (dashed outline: the undeformed reference planform). Top: open loop – bending and torsion coalesce and the response diverges within 0.93 s, at which point it leaves the linear regime and the rollout is stopped. Bottom: the identical pluck under centralised LQR, suppressed by the second frame and held near zero afterward. Agent colours match Figure 1.

strip (Jones, 1940). Control surfaces contribute quasi-steady thin-airfoil flap derivatives from the Glauert series, integrated over each surface's spanwise band. The equation of motion is

$$M\ddot{q} + C\dot{q} + Kq \; = \; L\,x_{\mathrm{w}} + B\delta + b_{\mathrm{g}}w_{\mathrm{g}}, \tag{2}$$

with plunge positive down and twist positive nose-up, following the sign convention shown in Figure 1b. The state has 56 dimensions: four modal coordinates and their rates, plus two wake states on each of 24 strips. Full geometric, structural and actuator parameters are tabulated in Appendix A for reproducibility.

3.5. Validation against the published benchmark

Table 2 compares the model against the Goland cantilever wing (Goland, 1945). Agreement is better than 0.5 % on the modal frequencies and the flutter speed, and this is enforced by an automated test against published values rather than against numbers the implementation once produced.

3.6. Cross-validation against an unsteady vortex-lattice solver

The model buys speed with two approximations a reader is entitled to distrust: two-dimensional strip aerodynamics, and quasi-steady flap derivatives that neglect the flap's own shed wake. We quantified both against SHARPy (del Carre et al., 2019), which couples an unsteady vortex-lattice

Table 2: Validation against the published Goland wing.

Quantity	This model	Published
First bending frequency	7.664 Hz	7.66 Hz
First torsion frequency	15.232 Hz	15.24 Hz
Flutter speed	$137.35\,\mathrm{m\,s^{-1}}$	$\approx 137\,\mathrm{m\,s^{-1}}$
Flutter frequency	$69.3\,\mathrm{rad\,s^{-1}}$	$\approx 70.7\,\mathrm{rad\,s^{-1}}$
Static divergence speed	$356.8\,\mathrm{m\,s^{-1}}$	—

method (Murua et al., 2012) to a geometrically exact beam. The comparison is clean because SHARPy's own Goland template carries structural data identical to ours, so any disagreement is about the aerodynamics.

Flutter point. SHARPy places flutter at $154.46\,\mathrm{m\,s^{-1}}$ and $69.3\,\mathrm{rad\,s^{-1}}$ at sea level, against $137.35\,\mathrm{m\,s^{-1}}$ and $69.34\,\mathrm{rad\,s^{-1}}$ for our model. The *frequency* agrees to 0.06 %, so both identify the same bending–torsion coalescence; the critical speed differs by −11.1 %. That direction is what two-dimensional aerodynamics predicts, since strip theory has no tip relief, over-predicts loading near the tip and therefore under-predicts the flutter speed. Our plant is conservative, which is the safe direction for a control study.

Control authority. This discrepancy runs the other way and is larger. A part-span flap loses effectiveness to spanwise carryover and to flow around its ends, neither of which a sectional derivative represents. Integrating our thin-airfoil $C_{l\delta} = 3.826$ per radian over the flapped span (25 % of span, read from the generated aerodynamic grid) gives a whole-wing $\mathrm{d}C_L/\mathrm{d}\delta$ of 0.957 per radian, against 0.495 from UVLM. Our plant over-estimates control authority by a factor of 1.9, grid-converged to within 3 % between 8×16 and 16×32 panels.

Consequence. Every controller in this study, classical and learned, was evaluated on the same plant and received the same inflated authority, so relative comparisons, ablations and significance tests are unaffected. Absolute decay rates would not transfer unchanged to a UVLM plant, and we do not claim they would. The two discrepancies act in opposite directions — a harder flutter problem, an easier control problem — so they cannot be combined into a single safety factor.

4. Method

4.1. An exactly additive energy budget

Differentiating the aeroelastic energy $E = \frac{1}{2}\dot{q}^\top M\dot{q} + \frac{1}{2}q^\top Kq$ along trajectories of (2) gives

$$\dot{E} = \underbrace{-\dot{q}^\top C\dot{q}}_{\text{dissipation}} + \underbrace{\dot{q}^\top L x_{\mathrm{w}}}_{\text{wake exchange}} + \sum_k \underbrace{(\dot{q}^\top b_k)\,\delta_k}_{P_k}, \tag{3}$$

with b_k the kth column of B. The final term is exactly additive across agents; $P_k < 0$ means surface k removes energy from the structure.

Proposition 1 (Closed-form difference reward)**.** *For a global objective whose only agent-dependent term is the control power in* (3)*, the difference reward* $D_k = G(z) - G(z_{-k})$ *admits the closed form* $D_k = -P_k$*. No counterfactual rollout, learned mixing network or centralised critic is required to evaluate it.*

Proof. Removing agent k sets $\delta_k = 0$. Every other term of (3) is independent of δ_k, and the control-power term is a sum over agents, so the difference collapses to $(\dot{q}^\top b_k)\delta_k$. □

Projecting onto the in-vacuo modes keeps the separation in agent *and* mode: agent k's contribution to mode r is $\dot{\eta}_r(v_r^\top b_k)\delta_k$. The factor $v_r^\top b_k$ is the modal residue, which decides whether a surface damps or excites that mode and whose sign inverts under control reversal.

4.2. The credit must be the logarithmic decay rate

Used directly, $-P_k$ fails instructively. It is maximised by having energy available to extract, so an agent is rewarded for *sustaining* the oscillation. Policies trained on it held the wing at a bounded but undamped amplitude, scoring $+0.265\,\mathrm{s}^{-1}$ against -0.06 for a hand-tuned collocated damper. This is reward hacking, not myopia.

Dividing (3) by E repairs it:

$$\frac{\mathrm{d}}{\mathrm{d}t}\log E = \frac{-\dot{q}^\top C\dot{q} + \dot{q}^\top L x_{\mathrm{w}} + \sum_k P_k}{E}, \qquad r_k = -\frac{P_k}{E + \varepsilon}. \tag{4}$$

Because E is a shared scalar the decomposition survives division exactly, so Proposition 1 carries over to the log-decay rate. The signal is now scale invariant, and it coincides with the evaluated objective (1). Making the change inverted the sign of the outcome, from $+0.265$ to $-0.291\,\mathrm{s}^{-1}$, with training divergence falling from $0.8\,\%$ to $0.0\,\%$.

Remark 1. *The exact signal remains myopic: cross-mode coupling through C and through the wake means an agent may have to inject energy into one mode so another can extract more from the coupled pair. We therefore add potential-based shaping (Ng et al., 1999) on $\log E$. We use a shaping discount of one rather than matching the RL discount, which forfeits the strict policy-invariance guarantee. The reason is concrete: with a discount below one the leakage term $(\gamma - 1)\Phi/\Delta t$ becomes a constant negative reward once the wing is damped — measured at -6.91 per step, totalling -977 over an episode for a controller that damps perfectly, against roughly zero for one that lets the wing ring. An invariance theorem is of no use if the shaped reward ranks a ringing wing above a damped one.*

4.3. Training

We use shared-parameter multi-agent PPO (Schulman et al., 2017; Yu et al., 2022) with generalised advantage estimation (Schulman et al., 2016) and per-agent advantages, since each agent receives a different reward. An on-policy method is preferred here over off-policy alternatives (Lillicrap et al., 2016; Haarnoja et al., 2018) because the plant is cheap to simulate, so sample efficiency is not the binding constraint, and because the per-agent credit of Section 4.2 is a function of the current state and action that is recomputed exactly at every step rather than stored and replayed. Updates run on sequences: minibatches are segments of consecutive steps unrolled with gradient from a detached stored hidden state. Sampling shuffled individual timesteps, as we did initially, trains any recurrence as a one-step map and gives a recurrent encoder no temporal gradient at all (Hausknecht and Stone, 2015). A single illustrative run of `ippo_physics` kept under the old shuffled-timestep sampler reached $-0.213\,\mathrm{s}^{-1}$, against -0.566 ± 0.123 for the same variant under the sequence-based sampler used everywhere else in this paper (Table 4); we report this as one run, not a claim, but it is the right order of magnitude for what a broken temporal gradient should cost. Algorithm 1 summarises the procedure.

4.4. Network architecture

The ablation in Section 5.7 names four architectural variants by the mechanism each adds; this section is what those names refer to, since a table of decay rates cannot substitute for knowing what `mocca` or `comm_raw` actually computes. Figure 3 shows the full pipeline, with the two optional stages — the encoder and the consensus rounds — and the optional action head shaded separately from the parts every variant shares.

Algorithm 1 Training with the blended (exact-plus-shaping) credit

Assemble M, C, K, L, B at the sampled airspeed; small-gain init of the action head
for each update **do**
  Ease the task distribution by the curriculum fraction
  for $t = 1, \ldots, T$ **do**
    Each agent k acts on its own o_k^t; step the coupled plant and actuators
    $P_k^t \leftarrow (\dot{q}^\top b_k)\,\delta_k^t$ ▷ closed form, Prop. 1
    $r_k^t \leftarrow -P_k^t/(E^t + \varepsilon) + w\,[\Phi^{t+1} - \Phi^t]/\Delta t$ ▷ $\Phi = -\log \frac{E+\varepsilon}{E_{\mathrm{ref}}}$
  end for
  Compute per-agent GAE advantages
  for each epoch, each minibatch of *sequences* **do**
    Unroll L_{bptt} steps from a detached hidden state; zero it where an episode ended
    PPO clipped objective + value loss − entropy bonus
  end for
end for

Shared trunk. Every variant feeds agent k's local observation $o_k^t \in \mathbb{R}^{12}$, concatenated with a one-hot agent identity and (when enabled) an incoming consensus message e_k^t, through a two-layer tanh multilayer perceptron of width 128 to a latent ℓ_k^t. Parameters are shared across agents; only the one-hot identity varies, which is what keeps the parameter count independent of the number of surfaces (Table B.9) and is what makes the eight-surface scaling probe in Section 5.11 possible without retraining a wider network.

Phasor encoder (`recurrent_only` and beyond). A single instantaneous accelerometer reading cannot distinguish a mode moving up from the same displacement moving down, so recovering phase needs memory. A per-agent GRU cell (Hausknecht and Stone, 2015) carries a hidden state $h_k^t \in \mathbb{R}^{64}$ across the episode:

$$h_k^t = \mathrm{GRU}\Big(o_k^t, h_k^{t-1}\Big), \qquad \hat{\phi}_k^t = \tanh\Big(W_{\mathrm{out}}\, h_k^t\Big) \in \mathbb{R}^{2R}, \tag{5}$$

read as R=2 pairs $(\hat{\phi}_{k,r}^{\cos}, \hat{\phi}_{k,r}^{\sin})$, one per retained mode. The output is passed through tanh before it is used anywhere downstream: an unbounded phasor multiplied by a gain inside the action head saturates that head's own tanh at initialisation and kills the gradient before training starts, which is what the first implementation of this policy did. The phasor is carried as a (cos, sin) pair rather than an angle throughout, so that wrap-around is not a discontinuity the network has to fight.

Phasor consensus (`mocca` and `comm_raw`). The communication mechanism is $T{=}2$ rounds of averaging over the spanwise line graph — agent k exchanges only with agents $k{-}1$ and $k{+}1$, which is what a distributed avionics bus physically offers. The operator

$$P = (1 - w)\, I + w\, A, \qquad w = 0.5, \tag{6}$$

with A the row-normalised adjacency of the line graph, is fixed rather than learned, so consensus adds no parameters; the bandwidth cost is the message itself, $2R{=}4$ numbers per agent per round regardless of how many surfaces there are. The consensus message is $e^t = P^T \hat{\phi}^t$ (T applications of P), read back into the trunk of each agent as e^t_k. `comm_raw` replaces $\hat{\phi}^t_k$ in this pipeline with the agent's raw local observation instead of the encoder's phasor estimate, testing whether the phasor abstraction itself is doing anything relative to simply broadcasting what a neighbour sees. When the health channel is enabled (used only in the residual-policy probes of Section 5.11), each agent's own saturation fraction and its magnitude are concatenated onto the message before consensus, since phase alone cannot tell an agent that a neighbour's actuator has stopped responding.

Phase-locked action head (the full `mocca` architecture). For retained mode r with consensus phasor $(\hat{\phi}^{\cos}_r, \hat{\phi}^{\sin}_r)$, the head emits a gain $A_r \in [-1, 1]$ and a phase offset via a unit vector $(\cos\psi_r, \sin\psi_r)$, and acts by rotating the phasor:

$$\rho_r = \hat{\phi}^{\cos}_r \cos\psi_r - \hat{\phi}^{\sin}_r \sin\psi_r, \qquad a_k = \tanh\left(u + \sum_{r=1}^{R} A_r\, \rho_r\right), \tag{7}$$

where u is an additional broadband channel from the same latent. The resonant term is *added* to u rather than replacing it, which makes (7) a strict generalisation of the plain head: setting every A_r to zero recovers an unconstrained policy exactly. An earlier version of this head replaced u instead of adding to it, which turns a resonant prior into a restriction, and that version scored worse than removing the head entirely — the version reported as `mocca_nohead` in Table 5 is the trunk and consensus without this head, i.e. with the plain tanh(MLP) output shown at the top of Figure 3.

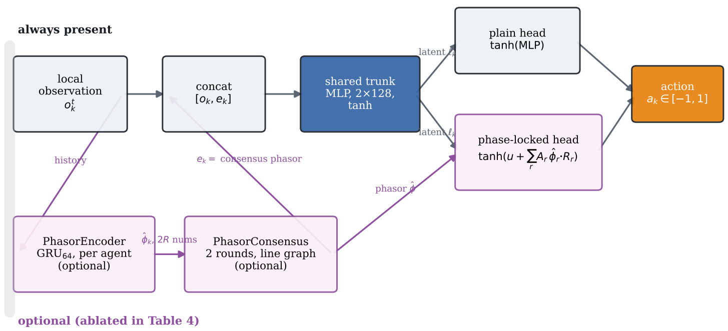

Figure 3: The policy network. Grey boxes and the plain head are present in every variant; the pink boxes and the phase-locked head are the mechanisms ablated in Table 5, so `recurrent_only` keeps the encoder but not consensus or the head, `mocca_nohead` keeps the encoder and consensus but ends in the plain head, and `mocca` is the full diagram.

Critic and exploration. The critic is a separate two-layer MLP that by default takes the same per-agent input as the actor; a fully centralised critic (fed the true modal state instead) was tried and made no measurable difference, consistent with this being a dense-reward, low-partial-observability credit problem rather than one where value estimation is the bottleneck, so we do not report it as a separate row. Every head, plain or phase-locked, has its final layer scaled by 0.01 at initialisation (Table B.9) so a fresh policy is close to a no-op, which is what makes the residual variants of Section 5.5 well-posed from the first update rather than starting by fighting the base controller. Actions are Gaussian with a learned, state-independent log-standard-deviation shared across agents — the quantity swept in Section 5.9.

5. Experimental evaluation

5.1. A baseline ladder that separates method from information

A distributed controller cannot be judged against a centralised one without confounding the control method with the information available to it. We report six rungs: collocated local rate feedback (no model, no communication); gain-scheduled LQR on the full 56-state plant including wake states, which is not implementable and serves as a ceiling; centralised LQG driven by a Kalman observer on the same six accelerometer channels the agents see; a fully decentralised LQG in which each surface runs its own observer on its own two accelerometers and commands only itself (Šiljak, 1991; Anderson and Moore, 2007); and the same decentralised design re-tuned for control-authority margin (Section 5.6). All are synthesised in discrete time at the control rate. The fully decentralised rung is the honest target.

5.2. Feasibility

We additionally run a fault-aware oracle: a gain bank re-synthesised per failure hypothesis, given the full state and told which surface failed. Its role is feasibility, not competition. Of nine evaluation conditions, 6 are stabilisable by this oracle; the rest are stabilisable by no controller with these actuators and are reported as infeasible rather than as failures of any method.

5.3. Metrics

PPO return cannot rank runs whose reward functions differ, so every controller is scored on the same physical quantities over identical episodes — same airspeeds, initial deformation, turbulence and injected failure: the fraction of episodes leaving the linear regime, the decay rate (1), peak tip plunge, and RMS deflection. Following (Agarwal et al., 2021) we report effect sizes and significance rather than means alone.

5.4. Training convergence

Figure 4 gives the training-time behaviour behind every learned number in this section: mean structural energy and divergence rate over training, evaluated on the curriculum task distribution rather than the held-out grid of Table 3, averaged across the seeds reported for each variant. Three things are worth stating plainly, since a final decay rate alone does not show them. First, every variant is within a small margin of its asymptote by 750 000 of the 1 500 000 environment steps budgeted (Table B.9); the training budget was not tuned to the point of convergence after the fact; it was chosen with headroom, and Figure 4 is the check that the headroom was enough. Second, `residual_only` starts substantially lower than every other variant and stays there for the first 200 000 steps, which is the training-time signature of the small-gain initialisation of Section 4.4: a residual policy that begins as a near-no-op inherits the decentralised LQG base law's performance from update one, rather than the near-open-loop performance every other variant starts from. Third, the divergence rate — which is a training statistic, not the evaluation metric of Table 3 — spikes above 0.5 % only in the first few thousand steps, before any policy has learned anything, while the curriculum's speed range and failure rate are still ramping up from their easiest

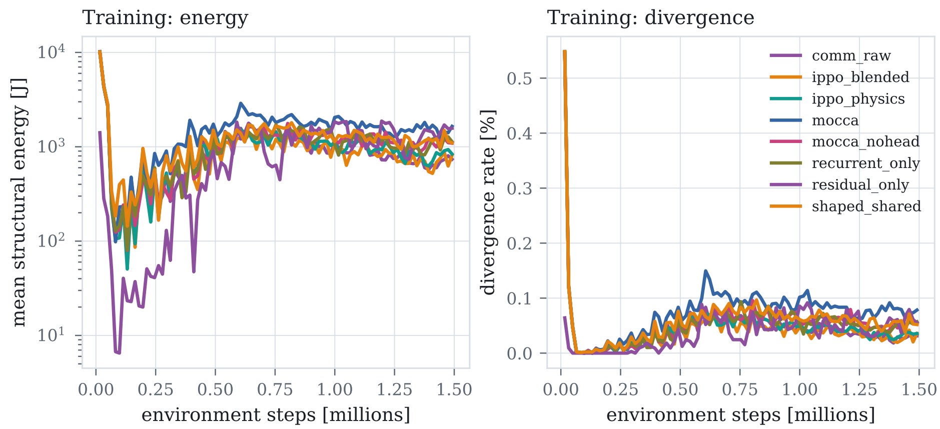

Figure 4: Training-time mean structural energy (left, log scale) and divergence rate (right) against environment steps, averaged over seeds, for every credit and architecture variant. All variants reach a stable plateau well inside the training budget; `residual_only` converges fastest because it starts from a decentralised LQG base law rather than from scratch.

setting (Algorithm 1; only the training *sampling* is eased, never the evaluation grid), and settles below 0.15 % for the remainder of training across every variant. We read this as evidence that the curriculum keeps training itself stable even where the resulting policy later struggles on the harder end of the evaluation grid; the two are different questions, and Figure 4 only answers the first one.

5.5. Results

Before the aggregate numbers, Figure 5 shows what a single trial they are computed from actually looks like: the same 5 cm pluck at $U/U_{\mathrm{f}} = 1.10$ under open loop, local rate feedback, and centralised LQR, sharing axes. Open loop grows into the divergence stop within 0.93 s. Local rate feedback neither grows nor decays — it holds a bounded, undamped oscillation rather than suppressing it, which is the signature of a controller that reacts to local motion without ever engaging the coupled flutter mode. Centralised LQR removes six orders of magnitude of structural energy within 1.5 s. Every entry in Table 3 is a decay rate fitted to a trace of exactly this shape, at one of nine flight and failure conditions.

Three things in Table 3 matter.

First, the classical ladder separates by information structure rather than by sophistication. Centralised LQG reaches $-5.52\,\mathrm{s}^{-1}$ on healthy conditions; the fully decentralised LQG, seeing exactly what the agents see, reaches

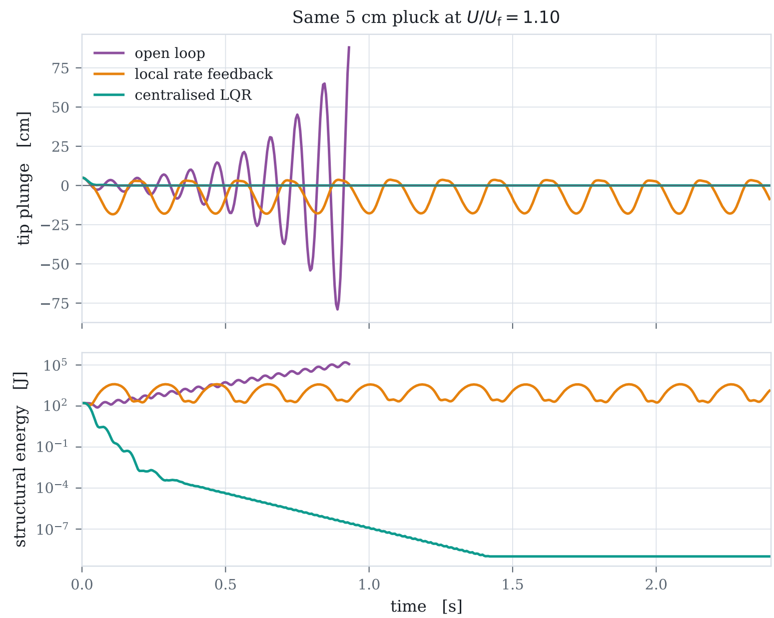

Figure 5: Tip plunge (top) and structural energy (bottom, log scale) for the same initial pluck under three of the five classical rungs. Local rate feedback neither grows nor decays: it damps local motion but never engages the unstable mode, which Section 5.8 confirms directly.

−2.05. Decentralisation costs a factor of 2.7 here, consistent with the classical expectation (Šiljak, 1991; Rotkowitz and Lall, 2006), and it is why the decentralised controller is the reference for a distributed policy.

Second, the best learned policy reaches $-1.53\,\mathrm{s}^{-1}$, about three quarters of the decentralised classical result. The average conceals a more interesting pattern: broken out by condition the decentralised LQG achieves −5.42, −1.07 and +0.32 at 1.05, 1.25 and 1.45 U_f — failing outright at the top of the envelope — against −1.98, −1.92 and −0.69 for the learned policy. The learned controller is far more uniform across the envelope and holds a condition where the classical one diverges, while losing badly where the classical one is strongest.

Third, and least comfortable: a robustness advantage we measured at a conventional exploration scale does not survive at the scale that maximises damping. At $\sigma = 0.30$ the learned policies held all six feasible conditions, matching the fault-aware oracle; at $\sigma = 0.05$ they hold five, as the classical controllers do, while damping three times better. Exploration scale trades

Table 3: Energy decay rate by controller and flight condition. Negative is suppressed, `D` marks the fraction of episodes that diverged, and `-` marks conditions infeasible for any controller.

Condition	local_fb	lqr_fixed	lqr_sched	lqg_local	dlqg_local	dlqg_robust	lqr_oracle	comm_raw	ippo_blended	ippo_physics	mocca_nohead	mocca	recurrent_only	residual_only	shaped_shared
U=1.05Uf healthy	-0.06	-4.40	-4.61	-4.71	-5.42	-4.17	-4.54	-0.78	-0.99	-0.90	-0.68	-0.85	-0.82	-1.98	-0.96
U=1.05Uf jam#2@8deg	-0.10	-0.10	-0.12	-0.13	-0.09	-0.11	-0.06	-0.05	-0.06	-0.06	-0.08	0.01	-0.08	-0.07	-0.10
U=1.05Uf jam#2@14deg	-0.12	-0.09	-0.10	-0.11	-0.08	-0.10	-0.06	-0.06	-0.08	-0.07	-0.06	D42%	-0.08	-0.07	D33%
U=1.25Uf healthy	-0.01	-5.27	-4.93	-5.38	-1.07	-0.82	-4.93	-0.62	-0.96	-0.77	-0.46	-0.47	-0.51	-1.92	-0.65
U=1.25Uf jam#2@8deg	D100%	D100%	D50%	D100%	D50%	D50%	-0.12	D75%	D100%	D100%	D100%	D100%	D100%	D50%	D100%
U=1.25Uf jam#2@14deg	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
U=1.45Uf healthy	D100%	-5.79	-5.68	-6.45	0.32	0.18	-5.74	-0.39	-1.30	-0.03	0.21	0.15	-0.09	-0.69	-0.47
U=1.45Uf jam#2@8deg	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
U=1.45Uf jam#2@14deg	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Stabilised	4/6	5/6	5/6	5/6	5/6	5/6	6/6	5/6	5/6	5/6	5/6	4/6	5/6	5/6	4/6

robustness against nominal performance here. We report both halves.

5.6. Is the classical baseline's fragility at the envelope edge a tuning artefact?

The comparison above rests on `dlqg_local` being a fair classical target, and an unstructured LQG has no guaranteed stability margin, unlike full-state LQR (Doyle, 1978) – so before reading anything into the learned policy holding a condition it diverges on, we checked whether a better-tuned classical design closes the gap, rather than asserting either way.

We first checked whether loop transfer recovery (Doyle and Stein, 1981) – the standard remedy for an *observer-induced* loss of the state-feedback loop's margin – helps: boosting the Kalman filter's process-noise weighting toward the full-state loop shape across four orders of magnitude changes the closed-loop control-authority margin at $1.45\,U_\mathrm{f}$ not at all (0.158 at every tested weighting; Appendix D). The reason is that the fully state-fed loop, with no observer at all, already sits at essentially the same margin: the observer is not where the fragility lives, so there is nothing for an observer-side remedy to recover. What does move the margin is re-weighting the state-feedback gain itself: a 50× increase in the control-effort weight ("expensive" control) raises the margin from 0.158 to 0.179 before plateauing, at a measurable cost to nominal decay rate (the closed-loop growth rate at design conditions worsens from 0.9744 to 0.9916 in discrete-time terms). That a fifty-fold retuning buys back so little control-authority margin, at a real performance cost, is itself the answer: the fragility at $1.45\,U_\mathrm{f}$ is structural to decentralised control at this operating point, not an artefact of the specific weights we happened to choose.

`dlqg_robust` in Table 3 is this re-tuned design, carried through the full nonlinear evaluation rather than only the linear margin analysis: it improves

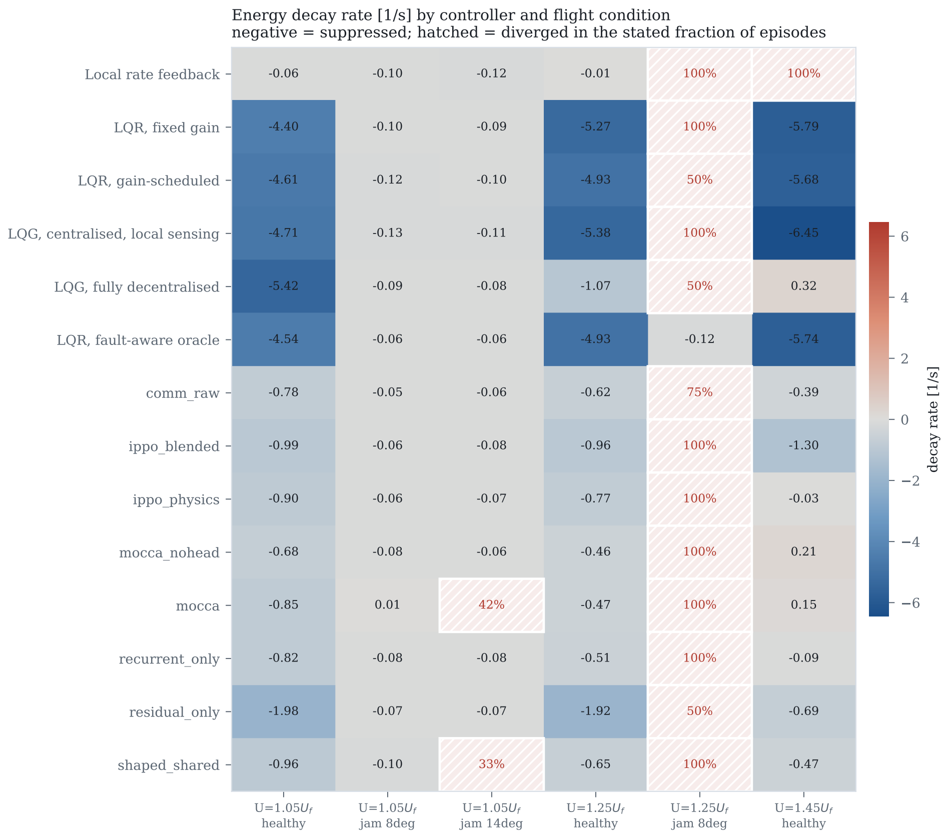

Figure 6: Decay rate by controller and condition. Hatching marks divergence.

the $1.45\,U_\mathrm{f}$ healthy-condition growth rate from $+0.32$ to $+0.18\,\mathrm{s}^{-1}$ – smaller, but still growing, still not a stabilised condition – while giving up nominal performance elsewhere (-0.82 against $-1.07\,\mathrm{s}^{-1}$ at $1.25\,U_\mathrm{f}$). Neither classical design holds the condition the learned policy holds. We also note, since the point was raised directly: the control-authority margin found above (0.158–0.179, i.e. stable down to 15.8 % to 17.9 % of the design authority) is comfortably clear of the plant's own measured control-authority discrepancy relative to UVLM (a $1.9\times$ overestimate, Section 3.6, i.e. true authority at $1/1.9 \approx 53\,\%$ of the design value) – a safety factor of roughly $3\times$ in relative-authority terms, in the direction that already occurred. The classical baseline's stability at the envelope edge is not close to being explained away by either an under-tuned weight choice or by the plant's own known modelling error.

5.7. Ablations

Credit assignment. Six seeds per variant at $\sigma = 0.05$ (Table 4), now five variants rather than three: the exact-plus-shaping combination (`ippo_blended`), each half alone (`ippo_physics`, `shaped_shared`), the naive undecomposed shared reward (`ippo_shared`, brought to the same six-seed standard from the single-seed probe an earlier version of this paper reported), and a learned value-decomposition credit in the style of VDN (Sunehag et al., 2018) (`vdn_shared`, Section 5.7.1) — the comparison against an actual estimated credit-factorisation method that Section 2 argues for but, in the version of this study reviewed previously, never ran. Reporting five variants together means ten pairwise comparisons rather than three, so every p-value in this paragraph and Table 4 is Holm-Bonferroni-corrected across that full family, and effect sizes are Hedges' g (small-sample-corrected) alongside Cohen's d. We also disclose, because a reviewer is entitled to ask how the six-seed standard was chosen rather than assume it was fixed in advance: the credit-versus-shaping comparison read positive, then null, then positive again across earlier, lower-seed sweeps before six seeds stabilised its sign, which is why this paper adopted a fixed seed count and a significance test in the first place rather than reading conclusions off small samples. That history does not retroactively invalidate the six-seed measurement, but it is part of why we treat the comparisons in this table as being at the edge of what six seeds can settle, rather than as more solid than the correction above already shows them to be.

The combination reaches -1.085 ± 0.243, and beats the closed-form credit alone (-0.566 ± 0.123) at $g = -2.49$, $p_{\mathrm{Holm}} = 0.020$ – the one comparison in this table that survives correction. It does *not* significantly beat shaping alone (-0.690 ± 0.145, $g = -1.82$, $p_{\mathrm{Holm}} = 0.080$) at this standard, which is a real retraction from the previous version of this paper: reported alongside only two other variants, that comparison's uncorrected $p = 0.009$ looked significant: it does not survive being tested as part of the five-variant family the physical question actually calls for. We do not think the mechanism we described – the closed form supplying an instantaneous term, the shaping supplying a long-horizon one – is wrong, only that this experiment does not establish it at the standard we now hold every comparison in this table to; a seed count larger than six, which is what the variance at six seeds leaves headroom for, is what would settle it.

The naive shared reward is not the outright failure a single seed originally suggested: at six seeds it averages -0.526 ± 0.806, damping well on some seeds (-1.969) and slightly exciting on others ($+0.060$), a spread three to six times wider than any credit-bearing variant's and indistinguishable from every one of them after correction ($p_{\mathrm{Holm}} \geq 0.57$ against all three). The honest statement is not that it fails, but that it is far less reliable: several times the run-to-run variance of the credit-bearing variants covers the range from competitive to mildly destabilising, so a single run either succeeding or failing tells a reader little about the method. The value-decomposition credit is the opposite failure mode: its own mean, -1.670 ± 0.699, is the most negative (best-damping) of all five and beats the closed-form credit alone at $g = -2.03$, $p_{\mathrm{Holm}} = 0.090$ – short of significance after correction, but the largest point estimate in the table – while its own variance is nearly three times `ippo_blended`'s, and it stabilises only 1.8 of nine conditions on average against 5.0 for every credit-bearing variant (Table 4): it learns to damp harder on the conditions it stabilises at all, at the cost of stabilising far fewer of them. We read this as the two failure modes a learned credit signal can have when the environment does not hand it one: no decomposition at all gives high-variance, unreliable performance; a decomposition estimated by summing per-agent value heads to a team target gives aggressive but inconsistent performance, holding fewer conditions than either the naive baseline or the closed-form credit. The physics-derived signal is not the only usable one, but it is the only one in this comparison whose seed-to-seed spread (0.123–0.243) and stabilised-condition count (5.0/9, matched by `ippo_physics` and `ippo_blended`) are both consistent across seeds, which is a property a practitioner should weigh alongside the mean.

5.7.1. A learned value-decomposition credit baseline

VDN (Sunehag et al., 2018) and QMIX (Rashid et al., 2018) solve a value-based, discrete-action credit-assignment problem: a joint action-value function is decomposed into per-agent utilities so that the greedy action for each agent can be read off independently. Our setting is continuous-action, on-policy actor-critic, so we adapt the decomposition rather than porting the algorithm unchanged: each agent keeps its own decentralised state-value head $V_i(o_i)$ (architecturally identical to every other variant's critic), but under the naive shared reward the summed value $V_{\mathrm{tot}} = \sum_i V_i$, not each V_i independently, is fit to the team return via generalised advantage estimation (Schulman et al., 2016) computed once at the team level, and every agent's policy update uses that shared team-level advantage. This is the natural on-policy analogue of value decomposition – it decomposes the *value function*, which is what VDN's architecture does – while being explicit that it does not reproduce VDN's original credit mechanism, which relies on each agent's utility depending on that agent's own discrete action, a property a shared state-value baseline does not have. Table 4 and the paragraph above give the result: better raw damping where it holds a condition, worse and more

Table 4: Credit-assignment ablation at $\sigma = 0.05$, six seeds per variant, five variants: the closed-form credit, log-energy shaping, their combination, a naive undecomposed shared reward, and a learned value-decomposition credit (Section 5.7.1). All ten pairwise p-values are Holm-Bonferroni corrected across this family.

Variant	Seeds	Decay rate [1/s]	Conditions held
`ippo_blended`	6	-1.085 ± 0.243	5.0/9
`ippo_physics`	6	-0.566 ± 0.123	5.0/9
`ippo_shared`	6	-0.526 ± 0.806	4.0/9
`shaped_shared`	6	-0.690 ± 0.145	4.3/9
`vdn_shared`	6	-1.670 ± 0.699	1.8/9

Comparison		Cohen's d	Hedges' g	p	p_{Holm}
`ippo_blended`	vs `ippo_physics`	-2.69	-2.49	0.0020	0.0198*
`ippo_blended`	vs `ippo_shared`	-0.94	-0.87	0.1557	0.5683
`ippo_blended`	vs `shaped_shared`	-1.97	-1.82	0.0088	0.0795
`ippo_blended`	vs `vdn_shared`	1.12	1.03	0.0996	0.4980
`ippo_physics`	vs `ippo_shared`	-0.07	-0.06	0.9078	1.0000
`ippo_physics`	vs `shaped_shared`	0.92	0.85	0.1421	0.5683
`ippo_physics`	vs `vdn_shared`	2.20	2.03	0.0112	0.0895
`ippo_shared`	vs `shaped_shared`	0.28	0.26	0.6426	1.0000
`ippo_shared`	vs `vdn_shared`	1.52	1.40	0.0257	0.1543
`shaped_shared`	vs `vdn_shared`	1.94	1.79	0.0177	0.1240

variable robustness overall than either the naive baseline or the closed-form credit. We read this as evidence that VDN-style decomposition, adapted this way, is not a free improvement over an undecomposed shared reward on this task, rather than as a demonstration that value decomposition cannot work here in some other form; COMA's counterfactual baseline (Foerster et al., 2018) and QMIX's monotonic mixing network (Rashid et al., 2018) were not adapted and tested, and we do not claim this one experiment closes the comparison the review that shaped this section asked for, only that it opens it honestly.

Architecture. The mechanisms we derived from the same physics do not survive measurement, and at six seeds per variant none of the four survives even marginally (Table 5). A plain feedforward policy with the blended credit reaches -1.085 ± 0.243; adding a recurrent phasor encoder gives -0.474 ± 0.249, adding phasor consensus -0.308 ± 0.169, adding the phase-locked action head (the full architecture) -0.389 ± 0.219, and sharing raw neighbour observations -0.596 ± 0.227. All four are significantly worse than

Table 5: Architecture ablation at $\sigma = 0.05$. Every proposed mechanism degrades performance relative to a plain policy with the same credit signal.

Variant	Seeds	Decay rate [1/s]	Conditions held
`comm_raw`	6	-0.596 ± 0.227	5.0/9
`mocca`	6	-0.389 ± 0.219	4.2/9
`mocca_nohead`	6	-0.308 ± 0.169	5.0/9
`recurrent_only`	6	-0.474 ± 0.249	5.0/9

Comparison		Cohen's d	Hedges' g	p	p_{Holm}
`comm_raw`	vs `mocca`	−0.93	−0.86	0.1391	0.6953
`comm_raw`	vs `mocca_nohead`	−1.44	−1.33	0.0336	0.2018
`comm_raw`	vs `recurrent_only`	−0.51	−0.47	0.3961	1.0000
`mocca`	vs `mocca_nohead`	−0.41	−0.38	0.4907	1.0000
`mocca`	vs `recurrent_only`	0.36	0.33	0.5444	1.0000
`mocca_nohead`	vs `recurrent_only`	0.78	0.72	0.2104	0.8414

the plain policy after Holm-Bonferroni correction across the ten pairwise comparisons among the five variants in this family (the four architecture variants plus the plain `ippo_blended` policy they are each measured against), reporting Hedges' g alongside Cohen's d to correct for the small-sample bias at $n = 6$ ($g = -2.29$, $p_{\text{Holm}} = 0.013$ for the recurrent encoder; $g = -3.43$, $p_{\text{Holm}} = 0.001$ for consensus; $g = -2.78$, $p_{\text{Holm}} = 0.004$ for the full architecture; $g = +1.92$, $p_{\text{Holm}} = 0.034$ for raw neighbour sharing). Unlike the credit comparison below, every one of these four survives the stricter accounting. Every elaboration we proposed made things worse, and correcting for the number of comparisons made the conclusion no less sharp.

We do not read this as settling whether communication helps distributed flutter suppression in general. Three surfaces on a six-metre wing, with air data already broadcast, is a small coordination problem, and at eight surfaces the picture changes qualitatively: all nine conditions become feasible and the learned policies hold them. It does settle that on this problem the credit signal was worth deriving from the physics and the message content was not.

Figure 7 puts every controller from Sections 5.5 and 5.7 on common axes. The pattern is visible at a glance in a way the tables cannot show as directly: the learned variants cluster with the classical rungs on the robustness axis (top) while forming a visibly separate, weaker band on the nominal-performance axis (bottom) — with one exception. The residual policy, which corrects a decentralised LQG rather than replacing it, is the

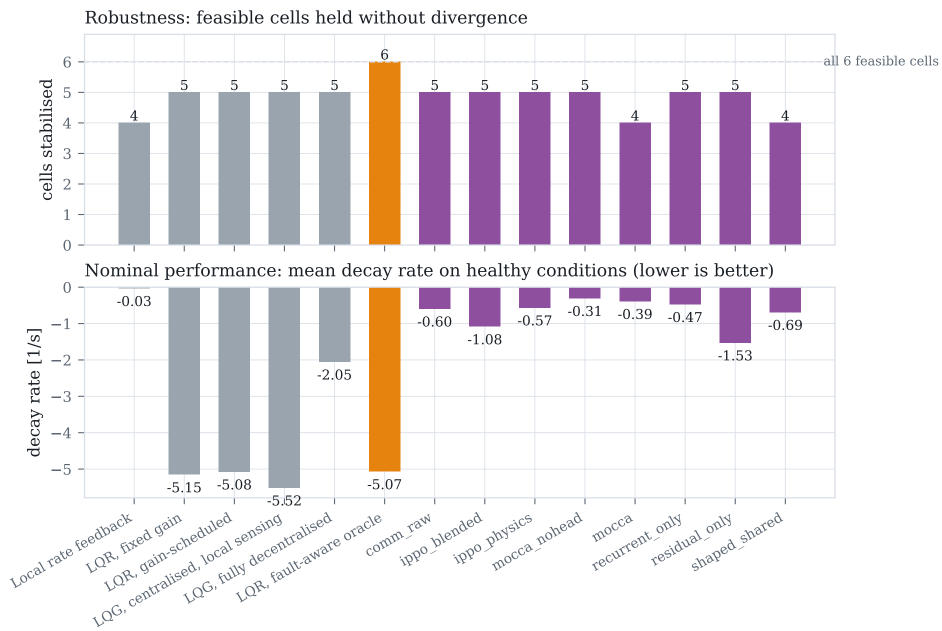

Figure 7: Every controller and variant on one axis. Top: feasible conditions held without divergence, out of the six the fault-aware oracle shows are stabilisable. Bottom: mean decay rate on healthy conditions. The learned variants (purple) match the classical rungs on robustness while trailing them on nominal damping, except for `residual_only`, which is the strongest learned result on both axes simultaneously.

only learned variant that reaches into the range the classical controllers occupy on *both* axes at once.

5.8. Emergent spanwise phase coordination

"Emergent coordination" is easy to assert about a multi-agent policy and hard to pin down, so before making any such claim we define it as something measurable: the phase at which each surface deflects relative to the critical structural mode's velocity, as a function of spanwise station. A controller with no notion of the mode shape should show no systematic phase progression along the span; one that has effectively discovered the mode shape should show a consistent progression matched to it. The measurement — the cross-spectrum of each surface's deflection against the first bending mode's velocity, evaluated at the frequency where the modal response peaks — is applied identically to every controller, classical or learned, so a learned pattern can be held against what optimal centralised synthesis produces rather than admired in isolation.

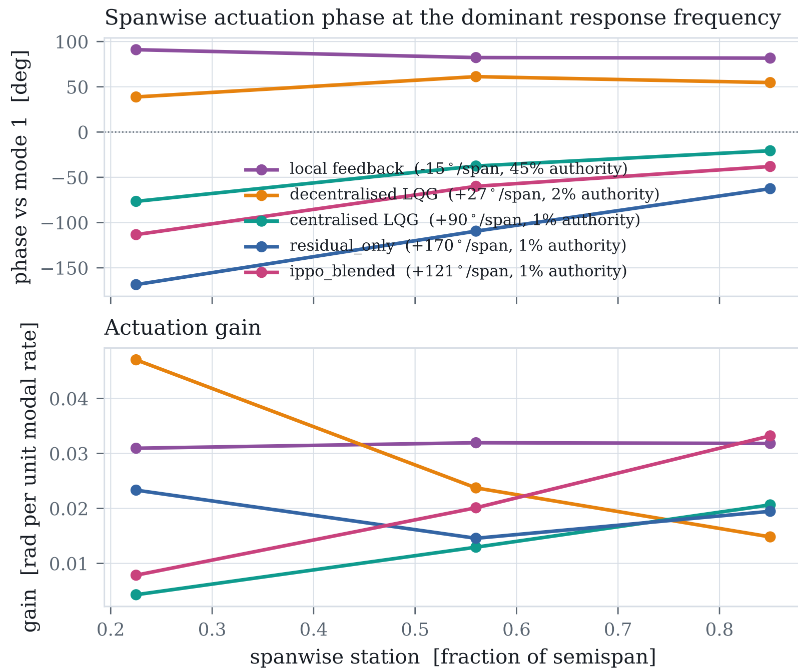

Figure 8: Spanwise actuation phase (top) and gain (bottom) at each controller's dominant response frequency. A sloped phase profile is a travelling-wave actuation pattern matched to the mode shape; a flat profile is purely local reaction. Legend entries give the phase gradient and the RMS actuation authority used.

Applied to the classical ladder the instrument already separates the rungs sharply (Figure 8, first three series). Local rate feedback is essentially flat at $-15°$ per semispan and responds at 5.50 Hz — it never engages the unstable mode at all, which is the mechanism behind its poor performance rather than a separate fact about it. The fully decentralised LQG reaches $+27°$ per semispan at 7.50 Hz, and centralised LQG develops $+90°$ per semispan while operating at 11.00 Hz — matching the 11.03 Hz the eigenvalue analysis of Section 3.6's plant predicts for the flutter mode almost exactly. The pattern is monotone: the more capable the controller, the stronger its spanwise phase progression, and only the optimal centralised design locks onto the true flutter frequency.

The learned policies give this a genuine test rather than a demonstration. The strongest credit-only policy, `ippo_blended`, locks onto 11.00 Hz — the same frequency as centralised LQG, and the same frequency the eigenvalue analysis predicts — with a phase gradient of $+121°$ per semispan, steeper than the centralised design's. It reaches this from six accelerometer channels, no communication, and no information about the mode shape beyond what the reward and dynamics implicitly encode. The residual policy locks onto a different, higher-frequency mode ($+170°$ per semispan at $15.50\,\mathrm{Hz}$) rather than the primary flutter mode, which is consistent with it correcting a base law that already suppresses the primary mode and spending its own authority on what is left. We read the first result as the concrete, falsifiable content behind the word "emergent" in this paper: a distributed policy that has never been told the mode shape can converge, from local measurement alone, on the same resonant frequency that full-state, full-model synthesis converges on by construction.

The same measurement also corrects a wrong inference authority alone would suggest. Local rate feedback uses $44.7\,\%$ of available deflection to reach $-0.03\,\mathrm{s}^{-1}$; centralised LQG uses $0.7\,\%$ to reach -5.52. The better the controller, the *less* authority it spends, because flutter suppression is a phase-precision problem rather than a control-power one. The learned policies ($1.2\,\%$ to $1.3\,\%$ authority) are accordingly not under-actuated, which a reading of the deflection magnitudes alone might suggest; they apply an amount of authority comparable to the good classical controllers, at a phase that is closer to optimal than local feedback achieves but not yet as sharp as the centralised design.

5.9. What limits learning here

Figure 9 is the finding with the widest reach, and the one we have now brought to the same six-seed standard as the ablations rather than the two seeds per point an earlier version of this paper reported. Holding everything else fixed and varying only the initial exploration standard deviation moves performance by a factor of two (-1.805 ± 0.257 at $\sigma = 0.05$ against -0.908 ± 0.271 at $\sigma = 0.15$, a ratio of 1.99), with an interior optimum at $\sigma = 0.05$ and a clear degradation above it – the qualitative shape an earlier, thinner sweep already suggested, and which six seeds confirms rather than overturns.

The mechanism can be read off the axis. A controller achieving the classical result uses about $0.105°$ RMS of deflection; $\sigma = 0.30$ corresponds to $4.5°$. The policy's own exploration is then some forty times the amplitude that works, and no credit decomposition can recover a signal buried underneath it.

This is not a tuning remark. Before we found it, every variant plateaued between -0.03 and $-0.82\,\mathrm{s}^{-1}$ regardless of credit signal, communication, action parameterisation, memory, auxiliary supervision or base controller. We had reported the seed-to-seed spread behind that plateau, from two

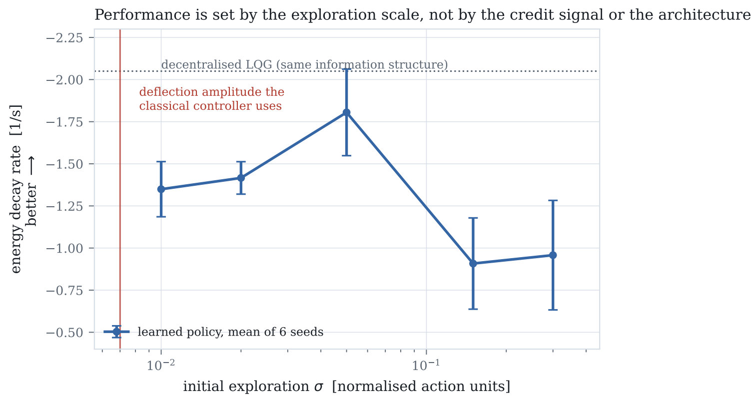

Figure 9: Decay rate against initial exploration standard deviation, all else fixed. The vertical line marks the deflection amplitude the classical controller uses. Six seeds per point.

seeds, as falling sharply between the conventional and matched scale (0.132 at $\sigma = 0.30$ to 0.016 at $\sigma = 0.05$); at six seeds it does not fall sharply at all (0.325 at $\sigma = 0.30$ to 0.257 at $\sigma = 0.05$), which we flag as a second instance in this paper of a two-seed spread understating the true variance, alongside the credit-versus-shaping retraction in Section 5.7. The interior optimum and the factor-of-two effect on the mean are what survive at six seeds; the earlier claim that the spread itself shrank sharply at the matched scale does not, and we withdraw it. A capability probe rules out the competing explanation: training on the single easiest condition gives +0.21 and +0.27, worse than training on the full distribution, so the plateau was never a matter of the task being too broad.

The practical recommendation is specific. The useful actuation amplitude is computable before any training, from a classical design or from an estimate of the damping authority required. Scale the exploration to it. On this problem that single number mattered more than every algorithmic choice we made, and it is cheap to check.

5.10. Sensitivity to control-influence identification error

The credit signal's sign is decided by the modal residue $v_r^\top b_k$ (Section 4), which is free in simulation but requires an identified control-influence estimate on hardware. A wrong sign rewards a surface for exciting a mode rather than damping it, so before recommending the signal for anything beyond simulation we quantify, rather than assert, how much identification error it tolerates.

Analytic: how large an error flips a sign. Modelling identification error as an independent relative perturbation on each $(mode, surface)$ entry of B, scaled to the natural magnitude of that mode's own row (entries in the same row share the same modal projection but each surface's own aerodynamic derivative is independently estimated in practice), the most fragile entry — the mid-span surface's (`aileron_mid`) residue on the first torsion mode — flips sign at just 11 % relative error, essentially identical across the flight envelope (11.0 % to 11.0 % at 1.05–1.45 U_{f}, since the fragility ratio is speed-invariant by construction). Every other entry is more robust, ranging up to entries that do not flip within ±100 % error at all. This is the honest answer to whether error in B threatens the credit signal: for the least robust surface–mode pairing, yes, at an identification error smaller than many real aerodynamic-derivative uncertainties; for the rest, no.

Under a *uniform* per-surface scale error instead — every entry of a surface's column moving by the same fraction, which is what a simple over-all effectiveness mis-estimate would look like — every sign flips simultaneously at exactly −100 % relative error (classical control reversal), independent of magnitude. The plant's own reported UVLM control-authority discrepancy (Section 3.6) is +90 %, an overestimate, which moves the assumed value *away* from zero rather than toward it: a safety factor of roughly 2–3× in relative-error terms from the boundary that would matter, in the direction that already occurred.

Empirical: training with a misinformed credit signal. We retrained the exact-plus-shaping credit variant with the true dynamics held fixed and only the credit signal's belief about control authority perturbed by a uniform ±60 %, three seeds each — a reduced but disclosed seed count, in keeping with this paper's own standard for probes that are not held to the six-seed bar. At the unperturbed baseline the healthy decay rate is -0.441 ± 0.330; believing 60 % more authority than is true gives -0.259 ± 0.217, and believing 60 % less gives -0.570 ± 0.131. Neither direction collapses performance or reverses its sign, and the spread across all three conditions — including the unperturbed one — is wide enough at three seeds that we do not read a monotone trend into the ordering. Both perturbations remain within the range where the uniform-scale analysis above predicts no sign reversal (±60 % is well short of the −100 % boundary), so this result should be read as evidence that moderate uniform identification error degrades gracefully rather than catastrophically, not as a test of what happens past that boundary, which we have not run.

Table 6: Additional probes, one or two seeds each, at $\sigma = 0.05$. Every healthy-decay value is listed per seed, not averaged, because two points do not estimate a standard deviation. -- marks a probe not run for that variant.

Probe	Variant	Healthy decay [1/s], per seed	Held
Do the negative mechanisms of Table 5 also hurt as a residual correction?			
residual + raw comm.	`residual_comm`	−0.697, −0.247	6/9, 6/9
residual + health-channel consensus	`residual_health`	−0.518, −0.066	5/9, 5/9
Does supervising the encoder on the true modal phasor help (Section 4.4)?			
supervised encoder, plain head	`aux_encoder`	−0.219, −0.029	5/9, 5/9
supervised encoder, residual	`aux_residual`	−0.431, −0.510	5/9, 5/9
Eight-surface scaling (cf. Section 5.7)			
consensus, plain head, N=8	`mocca_nohead`	−0.028, −0.015	9/9, 8/9
recurrent only, N=8	`recurrent_only`	+0.135, −0.011	9/9, 9/9

5.11. Additional probes

The results above are held to a fixed standard: six seeds, Welch's t-test, Cohen's d and Hedges' g, Holm-Bonferroni corrected. Three further questions arose directly out of those results and were checked, but only at one or two seeds each — not because the questions matter less, but because reaching the main standard for every question this study raised would have multiplied its cost several times over. We report them as probes rather than findings: Table 6 gives every seed individually rather than a mean, so that no reader mistakes a two-point spread for an estimated standard deviation.

The negative architecture result survives moving to a residual policy. The plain-policy ablation (Table 5) found that consensus and raw neighbour sharing hurt. Applying the same two mechanisms to a policy that corrects a decentralised LQG base law instead of replacing it — the setting where `residual_only` is the strongest result in the paper at $-1.53\,\mathrm{s}^{-1}$ (Section 5.5) — gives `residual_comm` at −0.697 and −0.247 and `residual_health` at −0.518 and −0.066: both clearly below `residual_only`, at both seeds. The negative result is not an artefact of the specific plain-policy setting it was first measured in.

Supervising the encoder on ground truth does not help. `aux_encoder` adds a loss term supervising the phasor encoder directly on the true modal phasor — privileged information, unavailable outside simulation — rather than letting the encoder's representation be shaped purely by the RL objective. Both seeds (-0.219, -0.029) sit below the un-supervised `mocca_nohead` result of -0.308 ± 0.169 (Table 5), and the residual counterpart `aux_residual` (-0.431, -0.510) sits far below `residual_only`'s -1.53. We read this as mild evidence that the credit signal already supplies what the encoder needs to represent, and that an additional supervised loss competes with, rather than complements, the policy gradient here — consistent with the broader pattern in Table 5 that mechanisms motivated by the physics did not survive contact with this specific plant and reward.

Eight surfaces changes what "held" means, not just how many. Section 5.7 reported that all nine evaluation conditions become feasible for the learned policies at eight surfaces, and Table 6 gives the numbers behind that sentence. They deserve a more careful reading than "held" alone provides: `mocca_nohead` holds nine of nine and eight of nine conditions, but at a healthy-condition decay rate of only -0.028 and -0.015 — an order of magnitude weaker than the three-surface result for the same architecture. `recurrent_only` holds every condition at both seeds, yet one seed's healthy-condition decay is *positive* ($+0.135$). "Held" in Table 6 means the episode did not cross the divergence threshold within its duration, not that the wing was driven to a decisively damped state; a bounded, barely-growing oscillation can hold a condition by this criterion without being a controller we would recommend. We stand by the qualitative claim that eight-surface architectures fail less catastrophically than three-surface ones, and retract any stronger reading of it than that.

6. Discussion and conclusion

6.1. Discussion

Pulling the pieces together: this study set out to ask whether treating each control surface as a separate learning agent is a workable way to suppress flutter, and the honest answer is conditional. On the credit-assignment question the answer is yes, with numbers behind it. On the question of whether the resulting policies are ready to fly, the answer is not yet, and the gap to a classical decentralised design is real rather than an artefact of tuning (Section 5.5). We think both halves of that answer are useful to a reader, which is why neither is relegated to a footnote.

For practitioners. The most actionable finding is the least glamorous one. Before training a policy for a physical control task, compute the actuation amplitude a competent classical controller would use, and check the initial exploration scale against it. On this plant the two were separated by roughly a factor of forty at a conventional PPO setting, and every algorithmic refinement we tried was invisible until that gap was closed. We would expect a version of this check to be worth running on any task where a linear or classical baseline is available even loosely, which covers most physical control problems a control engineering audience would consider.

Toward a certifiable architecture. Flutter suppression is a flight-safety-critical function by construction (Section 1): failure is loss of the aircraft, with no time for a pilot to intervene. Nothing in this paper argues that the learned policies as trained are ready for that role, and we say so explicitly. Every classical rung in Section 5 has a synthesis procedure with a known stability margin (Appendix D); a PPO-trained network has neither a certified margin nor, at present, a certification pathway, and we are not aware of one for any learned flutter-suppression policy in the literature we survey in Section 2, our own included. We place this work at approximately TRL 2–3 (technology concept formulated, analytical proof of concept): a validated simulation testbed and a credit signal derived from first principles, with no wind-tunnel or hardware-in-the-loop evidence yet.

The residual formulation offers the most concrete path we see to something closer to certifiable. `residual_only` (Section 5.5) is, by construction, a decentralised LQG base law — itself certifiable by the classical route — plus a learned correction initialised near zero (Section 4.4) and scaled by a fixed authority fraction. A run-time-assurance architecture built on the same structure — the certified base law as the default, the learned correction active only within a bounded authority fraction and a monitor that reverts to the base law if the correction's contribution or the plant's state leaves a pre-verified envelope — would let the learned component fail safe into a controller that is already certifiable on its own, rather than requiring the learned policy itself to carry a stability guarantee it does not have. We have not built or verified such a monitor; we flag it as the concrete next step this result points to; nor have we characterised how the credit signal's robustness to identification error (Section 5.10) interacts with the monitor's authority bound, which would need to be resolved before proposing this architecture to a certification authority.

For the credit-assignment literature. The result in Section 5.7 is, as far as we can tell, a genuine instance of a difference reward with no estimation step at all, on a task with real physical structure rather than a toy coordination game, and it beats the closed-form credit used alone at the standard we hold every comparison in this paper to. We had also hoped to report that it beats shaping used alone; at three variants and an uncorrected test it did, and that is what an earlier version of this paper claimed. Once compared as part of the five-variant family the physical question actually calls for – which is also what exposed that comparison's fragility – the shaping-alone comparison does not survive correction, and we retract the stronger claim rather than keep reporting a result that does not hold up to the standard the rest of this section is held to. Against an adapted value-decomposition baseline (Section 5.7.1), the closed-form credit's advantage is in reliability rather than raw mean: the learned decomposition reaches better damping on the conditions it stabilises, at the cost of stabilising fewer of them and with roughly triple the seed-to-seed variance. We think the more defensible summary of this section is narrower than our first submission's: the physics-derived signal is competitive with, not decisively better than, the alternatives we tested, and its main advantage is that it requires no estimation step and is consistent across seeds and conditions in a way none of the estimated alternatives were.

On the negative architecture result. We were not expecting the phasor consensus and phase-locked action head to hurt, and reporting that they do is worth more caution than reporting a positive result would need. A plausible interpretation is that a three-surface, six-metre wing with broadcast air data is simply too small a coordination problem for an explicit channel to earn its bandwidth cost in gradient-fitting difficulty, and the eight-surface result — where every condition becomes feasible for the learned policies, albeit with weak nominal damping (Section 5.11) — is at least consistent with that story, though we did not run it to the statistical standard we hold the main claims to. The residual-policy probes in Section 5.11 extend the same caution: neither raw neighbour sharing nor a health-carrying consensus message helps a residual correction any more than they helped the plain policy. We would read our result as a caution against building communication machinery into a distributed controller by default, on the strength of a physical argument, without first checking whether a plain policy with the same credit signal already does the job.

Why we still find phase coordination the most striking result. A policy trained end to end on a shaped scalar reward, using only its own two accelerometers, converges on the same resonant frequency that a full-state Kalman-filtered LQG converges on by construction (Section 5.8). That frequency is not supplied anywhere in the reward, the observation, or the network architecture; it is a property of the coupled structure that the policy has to find. We think this is the sense in which the word "emergent" is earned here, as distinct from being merely asserted.

6.2. Limitations

- The learned policies do not match the decentralised classical controller on average nominal damping (-1.53 against $-2.05\,\mathrm{s}^{-1}$). They are more uniform across the envelope and hold a condition where it diverges; we do not claim they dominate it. That the classical rung is genuinely, not just weakly-tuned, fragile at that condition is itself checked in Section 5.6 rather than assumed.

- The robustness result is scale-dependent, as reported in Section 5.5. We give the trade-off rather than its better half.

- The plant over-estimates control authority by a factor of 1.9 relative to UVLM (Section 3.6). Relative results are unaffected; absolute decay rates would not transfer. We have not retrained on the UVLM plant. The same simplification (independent, per-surface flap strips with no spanwise aerodynamic carryover between them) is what makes the credit decomposition of Proposition 1 exactly additive; we have not characterised how the decomposition would degrade under a higher-fidelity aerodynamic model with inter-surface coupling, only how much control-authority error the classical baseline's stability tolerates (Section 5.6) and how much identification error the credit signal's sign tolerates (Section 5.10).

- The credit signal requires the modal control influence $v_r^\top b_k$, free in simulation but requiring a modal model on hardware. Its sensitivity to identification error is characterised in Section 5.10, not asserted away: the most fragile surface–mode pairing tolerates only 11 % independent relative error before its sign reverses, though the plant's own measured discrepancy is well clear of that boundary.

- Proposition 1 concerns the difference reward, not optimality of the resulting policy.

- Ablations use six seeds for the credit comparison (now five variants, Section 5.7), six for the architecture comparison, and now six for the exploration-scale sweep (Section 5.9), all on a three-surface wing. The eight-surface results, and the three further questions in Section 5.11, are one- and two-seed probes and are not run to the same standard. Correcting for the number of pairwise comparisons made across the five-variant credit family (Holm-Bonferroni) changed which comparisons are significant, and bringing the exploration sweep to six seeds changed our own estimate of its seed-to-seed spread substantially, though not the qualitative shape of the curve; see Sections 5.7 and 5.9 for what did and did not survive.

- The exploration finding is established on this plant and this algorithm. We expect it to generalise to control tasks whose useful actuation band is narrow relative to the action range, but have not shown that.

- The learned policies have no certification pathway and no stability-margin guarantee, unlike every classical rung in the baseline ladder. We place this work at approximately TRL 2–3 and describe what we think the concrete next step would be, without having taken it, in Section 6.

6.3. Conclusion

Posing flutter suppression as a distributed learning problem exposes a credit structure the physics resolves in closed form: the difference reward is an agent's own control power, normalised by the instantaneous energy so that it coincides with the objective and cannot be gamed by sustaining the oscillation. Combining it with log-energy shaping beats the closed form used alone once compared, and corrected for multiple comparisons, against the full family of alternatives a reviewer would want to see it measured against; against estimated credit signals, its advantage is reliability across seeds and conditions rather than a uniformly larger effect.

The more transferable lesson is the caution. That result, and a clean negative result on the communication machinery, were both invisible until the exploration scale was matched to the actuation amplitude that damps the wing — a quantity computable in advance from a classical design. On this problem it mattered more than any algorithmic choice we made, and we suspect that is true of active vibration control more broadly.

CRediT authorship contribution statement

The authors contributed jointly to conceptualisation, methodology, software, validation, formal analysis, investigation, writing, and review.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Code and data availability

The plant, baselines, training code, evaluation harness and the SHARPy cross-validation scripts are available at the project repository, including `scripts/animate_flutter.py`, which regenerates the animations referenced in Section 3.4 for the four classical cases (open loop, centralised LQR, local rate feedback, and centralised LQR with the outboard surface jammed) at the flight condition used throughout the paper; `scripts/robust_baseline_check.py` and `aegis/analysis/decentralized_robustness.py`, which reproduce the control-authority margin analysis of Section 5.6; and `scripts/credit_sensitivity.py`, which reproduces the identification-error analysis of Section 5.10.

Appendix A. Testbed parameters

Table A.7 gives the geometric and structural parameters of the wing and Table A.8 the actuator and sensing parameters, common to every controller in this study. All are the values used to generate Table 2.

Appendix B. Training hyperparameters

Table B.9 lists the PPO and network hyperparameters shared across every learned variant in Sections 5.7 and 5.9, except where a hyperparameter is itself the subject of an ablation (the initial exploration standard deviation, and the presence or absence of the architectural mechanisms in Table 5). All runs share parameters across agents, with the agent index supplied as a one-hot input, so the parameter count does not grow with the number of surfaces.

Table A.7: Wing geometry and structural properties.

Parameter	Value
Semispan	6.096 m
Chord	1.8288 m
Mass per unit length	$35.71\,\mathrm{kg\,m^{-1}}$
Polar inertia per unit length (about EA)	8.64 kg m
Bending stiffness EI	$9.77 \times 10^6\,\mathrm{N\,m^2}$
Torsional stiffness GJ	$0.987 \times 10^6\,\mathrm{N\,m^2}$
Elastic axis	33 % chord
Centre of gravity	43 % chord
Bending / torsion modes retained	2 / 2
Aerodynamic strips	24
Air density (sea level)	$1.225\,\mathrm{kg\,m^{-3}}$

Appendix C. Compute and reproducibility

Every learned variant in this paper trains for $1.5{\times}10^6$ environment steps across 128 vectorised environments (Table B.9) on a single NVIDIA RTX 5000 Ada (laptop) GPU, using PyTorch 2.6. Across the 86 training runs behind the main results and ablations (Tables 3–5), median wall-clock training time is 331 s (minimum 246 s for a plain policy, maximum 1115 s for the full `mocca` architecture, whose GRU encoder and consensus rounds are the most expensive forward pass in the ablation set), at a median throughput of $4.5{\times}10^3$ environment steps per second — the batched environment of Section 5 rather than the network is the bottleneck at this scale, since the plant assembly in (2) is vectorised across environments but not across agents. The full set of runs reported in this paper — the credit and architecture ablations, the eight-surface scaling probe, and every additional probe in Section 5.11 — totals 12.7 h of GPU time; no result in this paper required infrastructure beyond a single workstation GPU.

Appendix D. Baseline controller synthesis

This appendix gives the synthesis equations behind every rung of the ladder in Section 5, so that “discrete-time LQG on local accelerometers” is reproducible from the paper alone rather than only from the repository.

Table A.8: Control surface, actuator and sensing parameters (identical for all three surfaces).

Parameter	Value
Spanwise bands (fraction of semispan)	$[0.05, 0.40]$, $[0.40, 0.72]$, $[0.72, 0.98]$
Hinge location	75 % chord
Deflection travel	$\pm 15^\circ$
Rate limit	$200\,^\circ\,\mathrm{s}^{-1}$
Actuator time constant	20 ms
Control update rate	200 Hz
Local observation per agent	2 accelerometers, plunge/twist rate, own saturation, air data

Appendix D.1. Augmented, discretised plant

Actuator lag is part of the design plant, not an unmodelled disturbance added afterwards: with $z = [q, \dot{q}, x_{\mathrm{w}}, \delta]$ (the state of (2) augmented with the deflections) and first-order servo lag of time constant τ_k per surface,

$$\dot{z} = \tilde{A}z + \tilde{B}u, \qquad \tilde{A} = \begin{bmatrix} A & B \\ 0 & -\mathrm{diag}(\tau_k^{-1}) \end{bmatrix}, \quad \tilde{B} = \begin{bmatrix} 0 \\ \mathrm{diag}(\tau_k^{-1}) \end{bmatrix}, \tag{D.1}$$

where A, B are the plant matrices of (2) in state-space form and $u \in [-1, 1]^K$ is the normalised command. Every controller in this study is synthesised in discrete time at the 200 Hz control rate via the exact zero-order-hold discretisation

$$\begin{bmatrix} \Phi & \Gamma \\ 0 & I \end{bmatrix} = \exp\left(\begin{bmatrix} \tilde{A} & \tilde{B} \\ 0 & 0 \end{bmatrix} \Delta t \right), \tag{D.2}$$

giving $z_{t+1} = \Phi z_t + \Gamma u_t$. Designing in continuous time and integrating the resulting gain with a forward-Euler step at $\Delta t = 5\,\mathrm{ms}$ is not a minor approximation here: the wake states have $|\lambda|$ up to $\sim 1.3 \times 10^4\,\mathrm{rad\,s^{-1}}$, so $\Delta t\,|\lambda| \approx 63$, and the observer built that way is unstable regardless of how good its gain is.

Appendix D.2. LQR gain

The state cost weights physical structural energy rather than an arbitrary diagonal, plus a light penalty on held deflection so the optimum does not

Table B.9: PPO and network hyperparameters.

Hyperparameter	Value
Parallel environments	128
Rollout length	128 steps
Epochs per update	4
Minibatches per epoch	4
BPTT segment length	32 steps
Learning rate (Adam)	3×10^{-4}
Discount γ	0.995
GAE λ	0.95
PPO clip range	0.2
Value loss coefficient	0.5
Entropy coefficient	2×10^{-3} (0 in the exploration sweep)
Max gradient norm	0.5
Curriculum fraction	0.4 of training
Hidden layer width (trunk)	128
Phasor encoder hidden width	64
Retained modes per phasor	2
Consensus rounds	2
Action-head output-layer gain (init.)	0.01

park surfaces at the travel limit:

$$Q = \begin{bmatrix} K_{\text{modal}} & & \\ & M_{\text{modal}} & \\ & & \operatorname{diag}\Big(w_{\text{rate}}/\delta^2_{\max,k}\Big) \end{bmatrix}, \qquad R = \operatorname{diag}\Big(w_{\text{effort}}/\delta^2_{\max,k}\Big), \tag{D.3}$$

with $w_{\text{effort}} = 2{\times}10^4$, $w_{\text{rate}} = 10^2$ throughout. The discrete-time algebraic Riccati equation

$$P = \Phi^\top P\Phi - \Phi^\top P\Gamma\left(R + \Gamma^\top P\Gamma\right)^{-1}\Gamma^\top P\Phi + Q \tag{D.4}$$

gives the gain $\mathcal{K} = (R + \Gamma^\top P\Gamma)^{-1}\Gamma^\top P\Phi$ and command $u_t = -\mathcal{K}z_t$ (Anderson and Moore, 2007). `BatchedLQR` solves (D.4) once at a fixed design speed; `BatchedScheduledLQR` solves it on a 9-point airspeed grid over the evaluation envelope and linearly interpolates $\mathcal{K}$ between grid points; `BatchedJamAwareLQR` additionally zeros the failed surface's column of Γ per failure hypothesis and resolves (D.4) for each hypothesis on the same grid, so the surviving surfaces' gains already account for the loss rather than compensating for it reactively.

Appendix D.3. Local rate feedback

The floor of the ladder uses no plant model: with $\dot{h}_k, \dot{\alpha}_k$ the locally sensed plunge and twist rate at surface k's own station,

$$u_k = -\left(g_h\, \dot{h}_k + g_\alpha\, \dot{\alpha}_k\right), \qquad g_h = 0.35,\ g_\alpha = 0.9, \tag{D.5}$$

clipped to $[-1, 1]$.

Appendix D.4. Steady-state Kalman observer on local accelerometers

The centralised and decentralised LQG rungs replace the true state z_t in $u_t = -\mathcal{K}\hat{z}_t$ with an estimate driven only by the accelerometer channels the agents themselves observe, $y_t = Hz_t$, where H is built from the acceleration rows of the plant rather than a selection matrix, since an accelerometer measures $\ddot{q}$, an algebraic function of the state rather than a state itself. The steady-state predictor-form filter is

$$\Sigma = \Phi\Sigma\Phi^\top + W - \Phi\Sigma H^\top \left(H\Sigma H^\top + V\right)^{-1} H\Sigma\Phi^\top, \tag{D.6}$$

$$L = \Phi\Sigma H^\top \left(H\Sigma H^\top + V\right)^{-1}, \tag{D.7}$$

$$\hat{z}_{t+1} = \Phi\hat{z}_t + \Gamma u_t + L\left(y_t - H\hat{z}_t\right), \tag{D.8}$$

with process noise $W = 10^{-4} I$ and measurement noise $V = 10^{-6}\,\mathrm{diag}(\|H_i\|^2)$ scaled per-channel, since accelerometer rows of H have norms of order 10^4–10^5 and an unscaled V demands impossible measurement precision and drives L to blow up (Anderson and Moore, 2007). `BatchedLQG` solves (D.6)–(D.8) once with H built from all six channels and $\mathcal{K}$ commanding all three surfaces jointly, on the same speed grid as the scheduled LQR. `Batched-DecentralizedLQG` solves an *independent* instance of (D.6)–(D.8) per surface k, with H restricted to k's own two accelerometer rows and Γ restricted to k's own column, so agent k's estimate, gain and command never depend on any other surface's measurement or actuation — the discrete-time, model-based analogue of the information structure the learned policies are given. Its scheduling grid spans the same $[1.0, 1.5]\, U_\mathrm{f}$ range as the training curriculum in Table B.9 exactly, by construction rather than coincidence, so the comparison in Section 5.5 is not confounded by the classical design being scheduled over a narrower envelope than the learned policy was trained on.

Appendix D.5. Re-tuning the decentralised LQG for control-authority margin

`BatchedRobustDecentralizedLQG` (Section 5.6) is structurally identical to `BatchedDecentralizedLQG`: same per-surface observer, same restriction to each agent's own two accelerometers and own actuator. Two remedies for its lack of a guaranteed margin (Doyle, 1978) were checked. The first, loop transfer recovery (Doyle and Stein, 1981), boosts the process-noise weighting W in (D.6) by $\rho\, b_k b_k^\top$ (agent k's own reduced input column) as $\rho \to \infty$, pushing the filter loop toward the full-state loop shape. Sweeping ρ from 0 to 10^4 changes the control-authority margin (the largest reduction in true control authority, relative to the design assumption, the closed loop remains stable under, found by bisection on the discrete-time closed-loop spectral radius) not at all: it stays at 0.158 at $1.45\, U_\mathrm{f}$ throughout. Comparing against the same gain applied with perfect (observer-free) state feedback, whose margin is also 0.158, shows why: the observer is not the bottleneck, so there is nothing for an observer-side remedy to recover.

The second remedy, and the one `BatchedRobustDecentralizedLQG` actually uses, re-weights the LQR cost (D.3) itself: w_effort increased from 2×10^4 to 10^6, with w_rate, W and V unchanged. This raises the control-authority margin from 0.158 to 0.179 before plateauing (no further gain to $w_\mathrm{effort} = 10^7$), at a cost to nominal decay rate documented in Section 5.6. `scripts/robust_baseline_check.py` reproduces both checks.

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